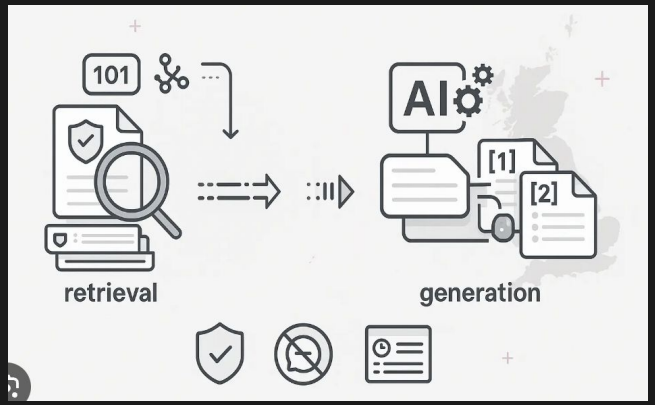
Retrieval-augmented generation for generative artificial intelligence in healthcare

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Summary

- Generative AI holds a lot of potential in the healthcare industry to enhance the quality of patient care and reduce clinic workload.
- However, we have to acknowledge the limitations of generative AI models. Includes being prone to biases and LLMs are unable to provide up-to date clinical advices.
- To overcome these challenges, retrieval-augmented generation (RAG) is a possible solution.
- RAG can help with bias reduction, disparity mitigation, mistake alleviation, health management



What is Retrieval-augmented Generation (RAG) ?

1) Indexing

- External data is split into chunks, encoded into vectors, and stored in a vector database

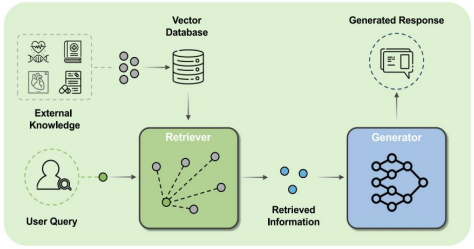
2) Retrieval

- Query gets encoded into a vector representation and the most relevant information is retrieved through similarity calculations between the query and information in vector database.

3) Generation

- Query and retrieved data are prompted by the model to generate information

Fig. 1 | A typical retrieval-augmented generation framework. External data is first encoded into vectors and stored in the vector database (where vectors are mathematical representations of various types of data in a high-dimensional space). In the retrieval stage, when receiving a user query, the retriever searches for the most relevant information from the vector database. In the generation stage, both the user's query and the retrieved information are used to prompt the model to generate content.



Application of RAG in Healthcare

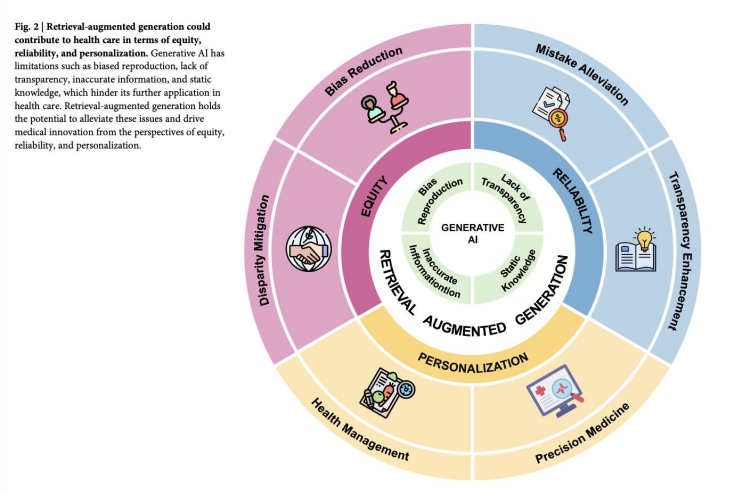


Fig. 2 | Retrieval-augmented generation could contribute to health care in terms of equity, reliability, and personalization. Generative AI has limitations such as biased reproduction, lack of transparency, inaccurate information, and static knowledge, which hinder its further application in health care. Retrieval-augmented generation holds the potential to alleviate these issues and drive medical innovation from the perspectives of equity, reliability, and personalization.

- RAG application in healthcare in these aspects: equity, reliability, and personalization
- Limitation of RAG in medical application scenarios and within the health system

Health Equity

- Bias Reduction:
  - RAG can help reduce demographic, gender, and racial biases
  - Retrieve targeted external medical literature, patterns, and clinical guidelines

- Disparity Mitigation:
  - RAG can help with healthcare access gaps in low-income/ resource and underrepresented communities
  - Leverage local data, digitalize regional guidelines, and have more multilingual knowledge



Reliable Content

- Mistake alleviation:
  - RAG can decrease medication and diagnostic errors by parsing through prescription at data entry stage
  - Use multimodal features for accurate drug identification

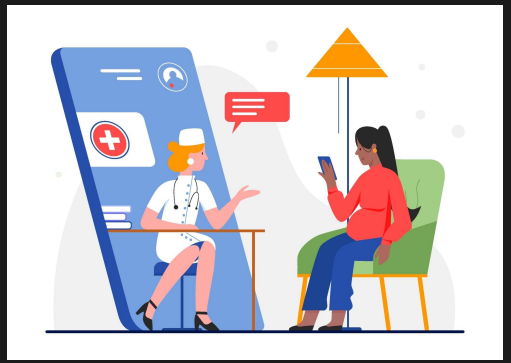
- Transparency enhancement:
  - RAG can improve transparency and trust by retrieving medical facts that can be tracked/logged, find evidence citations



Personalizing Healthcare Services

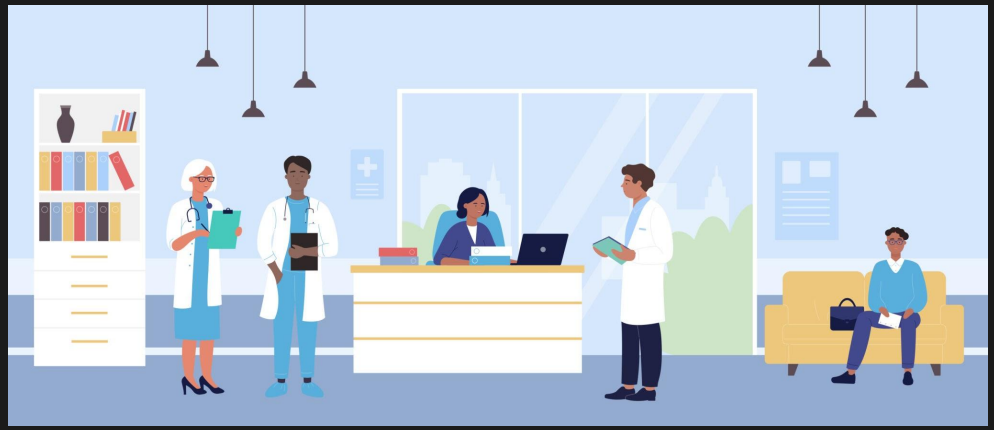
- Health Management:
  - RAG utilizes real-time electronic health records and individual lifestyle data to build unique profiles that is tailored towards a patient's disease prevention and medication

- Precision Medicine:
  - RAG can leverage biomarkers and clinical data to help with patient classification for personalized treatment plans



Limitations and Challenges

- RAG face practical challenges:
  - Biases from sources
  - Data scarcity for minority groups
  - Knowledge conflict
  - Privacy risks for patient data



Citations:

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Thank You!